

PROPOSAL RESEARCH

(Submitted under the Joint Call for Proposals 2025)

RIIM Kolaborasi Internasional – The 1st Joint Research Program
Between the Ministry of Science and Technology (MOST) of China
And the National Research and Innovation Agency (BRIN) of Indonesia

Project Title:

Thermohaline-convection-enhanced Solar Desalination coupled with Evaporative Crystallizer for Freshwater-Salt Coproduction under Tropical Climate Conditions

(Project Acronym: TROPIC-SALT)

Submitting Side: Indonesia

Principal Investigator (Indonesia):

Prof. Nasruddin, Universitas Indonesia, nasruddin@eng.ui.ac.id

Principal Investigator (China):

Prof. Zhenyuan Xu, Shanghai Jiao Tong University, xuzhy@sjtu.edu.cn

Research Theme:

A. Food and Agriculture	[Food biotechnology]	[Dairy products]
B. <u>New and Renewable Energy</u>	<u>[Solar energy]</u>	[Biomass and bio-fuels]
C. Transportation Technology	[Autonomous electric vehicles]	[Charging station]

(Please indicate your selected sub-theme(s) by using bold, underline, and blue font color.)

Date of Submission:

[20 August 2025]

Submitted to:

National Research and Innovation Agency (BRIN), Indonesia
(Indonesian-side submission)

Year 2025

APPROVAL SHEET

Joint Call for Proposals 2025
Between MOST (China) and BRIN (Indonesia)

*This form is to be completed and signed by the **Indonesian Principal Investigator** and their affiliated institution. It serves as an official endorsement of the proposal submission to BRIN under the Joint Call for Proposals 2025.*

1. Research Project Title:

Thermohaline-convection-enhanced Solar Desalination coupled with Evaporative Crystallizer for Freshwater-Salt Coproduction under Tropical Climate Conditions

(Project Acronym: TROPIC-SALT)

2. Principal Investigator (Indonesia):

a. Full Name (with academic title): Prof. Dr-Ing.Ir. Nasruddin, M. Eng

b. Affiliated Institution: Universitas Indonesia

c. Faculty / Department / Research Center & Research Organization: Mechanical Engineering, Faculty of Engineering, Universitas Indonesia

d. Institutional Address: Faculty of Engineering, Kampus UI, Kukusan, Beji, Depok City, West Java 16424

e. Phone Number: +62 815-8593-5355

f. Email: nasruddin@eng.ui.ac.id

3. Research Team Members (Indonesia):

No.	Name	Institution
1	Dr. Arnas Lubis	Universitas Indonesia
2	Erdiyanto Munandar	Universitas Indonesia
3	Mochamad Yusuf Efendi	Universitas Indonesia
4	Agus Nurbillah	Universitas Indonesia

4. Project Duration: From: [January/2026] To: [January/2028] (Duration: [24] months)

5. Total Requested Funding (Indonesia): IDR 1.988.900.000

6. Research Theme:

(Please indicate your selected sub-theme(s) by using bold, underline, and blue font color)

A. Food and Agriculture [Food biotechnology, Dairy products]

B. **New and Renewable Energy** [**Solar energy**, Biomass and biofuel]

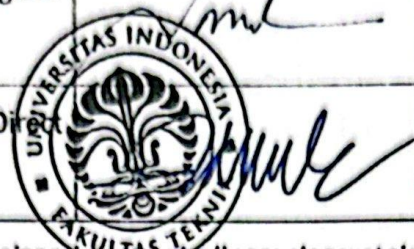
C. Transportation Technology (Autonomous electric vehicles, Charging station)

7. Partner Information (China Side):

Name of Principal Investigator (China): Prof. Zhenyuan Xu

Institution: Shanghai Jiao Tong University

8 Authorized Signatures - Indonesia Side:

Role	Signature & Stamp	Name & Title
Principal Investigator (Indonesia)		Prof. Dr-Ing.Ir. Nasruddin, M. Eng
Authorized Official / Director Supervisor of the PI		Prof. Kemas Ridwan Kurniawan, S.T., M.Sc., Ph.D.

Note: This form may be signed electronically or using wet signature. In case of mixed signature types, each party must ensure document authenticity and content consistency.

9. Acknowledged by Partner PI (China Side):

This proposal has been jointly prepared with the undersigned, who confirms awareness of its content and submission to BRIN.

Role	Signature & Stamp	Name & Title
Principal Investigator (China)		Prof. Zhenyuan Xu

Note: This signature confirms awareness of the proposal submission.

PROPOSAL TEMPLATE

RIIM Kolaborasi Internasional – BRIN and MOST

Joint Call for Proposals 2025

(The 1st Joint Research Program)

Between

the Ministry of Science and Technology (MOST) of China and the National Research and Innovation Agency (BRIN) of Indonesia

1. PROJECT TITLE

Thermohaline-convection-enhanced Solar Desalination coupled with Evaporative Crystallizer for Freshwater-Salt Coproduction under Tropical Climate Conditions

(Project Acronym: TROPIC-SALT).

2. PRINCIPAL INVESTIGATORS AND TEAM MEMBERS

Principal Investigator (Indonesia):

- Prof. Nasruddin, Universitas Indonesia, nasruddin@eng.ui.ac.id

Principal Investigator (China):

- Prof. Zhenyuan Xu, Shanghai Jiao Tong University, xuzhy@sjtu.edu.cn

No.	Name	Institution	Role in Project
1	Dr. Arnas Lubis	Universitas Indonesia	Solar Thermal System
2	Erdiyanto Munandar	Universitas Indonesia	System and CFD Simulation
3	Mochamad Yusuf Efendi	Universitas Indonesia	Renewable Energy, Salt, Biomass valorization
4	Agus Nurbillah	Universitas Indonesia	Material Science

No.	Name	Institution	Role in Project
1	Jie Yu	Shanghai Jiao Tong University	Crystallization Design
2	Renpeng Li	Shanghai Jiao Tong University	Double cell design

3. EXECUTIVE SUMMARY

Freshwater scarcity and the sustainable management of desalination brine are critical global

challenges. Conventional solar desalination technologies are often hindered by low efficiency and severe salt fouling, issues that are exacerbated in tropical climates characterized by high ambient humidity, fluctuating solar insolation, and a high propensity for biofouling. This proposal introduces the TROPIC-SALT project, an innovative system designed for the simultaneous production of freshwater and salt. The project aims to develop and validate a system integrating thermohaline-convection-enhanced solar desalination with a coupled crystallizer, specifically adapted for the unique environmental conditions of the tropics. The research methodology is structured around four key tasks: (1) developing a novel double-cell desalination unit for directional brine discharge; (2) designing and optimizing a multi-stage system for efficient hypersaline brine production; (3) integrating a crystallizer driven by waste condensation heat for salt recovery; and (4) creating an adaptive design for robust, long-term operation under real-world tropical weather. This project represents a strategic collaboration with the technology's originators at Shanghai Jiao Tong University (SJTU), with the Indonesian team leading the adaptation, fabrication, and rigorous testing of the integrated system. The TROPIC-SALT project aims to deliver a robust, low-cost, and decentralized solar-powered solution for water and salt security in coastal and archipelagic regions.

4. RESEARCH THEME ALIGNMENT

Selected Theme: New and Renewable Energy – Solar Energy

This project is fundamentally centered on the innovative application of solar energy, a primary renewable resource, to address critical water and resource management challenges. The core of the proposed TROPIC-SALT system is the highly efficient solar-thermal energy conversion process. The research focuses on optimizing every stage of solar energy utilization: from maximizing absorption and minimizing thermal losses via advanced system design, to recycling the latent heat of vaporization through a multi-stage architecture. This directly addresses the call's objective to develop and apply new and renewable energy technologies. By creating a decentralized, solar-powered solution, this project aligns with Indonesia's national priority to increase its renewable energy portfolio and contributes to the global agenda for clean energy and water security (SDG 7 and SDG 6).

5. BACKGROUND AND RATIONALE

Solar-driven desalination is a promising pathway to address global water scarcity, yet it faces two persistent challenges: energy efficiency and salt fouling. Conventional interfacial solar evaporators predominantly rely on porous wick structures to supply saline water to a photothermal surface. While effective for thermal localization, these wick-based systems are highly susceptible to pore clogging by salt crystals, especially when treating high-salinity brines. This fouling mechanism drastically reduces evaporation rates, degrades long-term performance, and limits the operational reliability of the devices [1], [2].

The Indonesian Context: A Dual Crisis of Water and Salt

The urgency of this research is profoundly amplified by Indonesia's specific national

challenges. As the world's largest archipelagic state, Indonesia faces a dual crisis: escalating water scarcity in its vulnerable coastal and small island communities, and a chronic dependency on industrial salt imports despite having the second-longest coastline globally. Projections indicate that the area of water-critical regions in Indonesia will increase from 6% in 2000 to 9.6% by 2045, affecting populous islands like Java, Bali, and parts of Sumatra and Sulawesi [3]. These regions are often afflicted by seawater intrusion into groundwater aquifers, rendering conventional water sources unusable.

Simultaneously, Indonesia faces a "salt paradox." The nation imports an average of 2.72 million tons of salt annually [4], primarily to meet the high-purity ($\text{NaCl} > 97\%$) requirements of its chemical and industrial sectors. Domestic production, largely from traditional solar evaporation methods, yields lower-quality salt that is unsuitable for these applications and is highly vulnerable to climate fluctuations [5]. This dependency creates significant economic losses and strategic vulnerabilities. This project directly addresses this dual crisis by proposing a technology that not only provides a climate-independent source of freshwater but also treats the desalination brine—typically a waste product—as a valuable resource for producing the high-grade industrial salt that Indonesia currently imports.

Technological Foundation and Partnership Synergy

Pioneering research by our esteemed partners at Shanghai Jiao Tong University (SJTU) has led to fundamental breakthroughs that overcome these limitations. By replacing the conventional wick with a confined saline layer in a tilted, multi-stage device, they initiated strong thermohaline convection. This natural circulation, driven by coupled temperature and salinity gradients, effectively rejects salt from the evaporation interface back into the bulk fluid. This mechanism confers extreme salt resistance, enabling stable, continuous operation with near-saturated brine (20 wt%) and achieving record-high solar-to-water efficiencies (up to 385% in a 10-stage device) through latent heat recovery [6], [1]. Furthermore, their research has paradigmatically shifted the view of salt crystals from a problem to a solution. They have shown that salt crystals can be manipulated to self-assemble into a hierarchical porous structure that functions as an ideal evaporator. This "Hierarchical Porous Salt Evaporator" (HPSE) enables extremely high evaporation rates from hypersaline brine, facilitating an effective pathway towards Zero Liquid Discharge (ZLD) [7].

This partnership is exceptionally synergistic. The SJTU team, led by Prof. Xu, are world-leading experts in the fundamental physics and modeling of these systems. The Indonesian team will provide the critical contribution of adapting, fabricating, and validating this technology under real-world tropical climate conditions—an environment where its performance has not yet been explored. The Indonesian team, led by Prof. Nasruddin, is exceptionally well-positioned for this role, possessing a robust track record in renewable energy research and development. Prof. Nasruddin's research portfolio directly aligns with the objectives of the TROPIC-SALT project. His work includes comprehensive performance analyses of solar thermal systems, such as evacuated tubular collectors for cooling applications [8], and extensive investigations into integrated multigeneration systems powered by renewable sources [9], [10]. Notably, his research has explored the techno-economic feasibility of polygeneration systems designed to

produce electricity, freshwater via desalination, and cooling, demonstrating a clear focus on developing integrated energy solutions to meet essential community needs [9]. Specifically, one of his current research focuses on developing predictive models and optimizing integrated processes for freshwater production and salt crystallization from seawater, which provides a highly relevant foundation of expertise for this project. This ongoing research specifically investigates evaporation-condensation dynamics under humid tropical conditions and their influence on salt crystallization kinetics. The objective is to maximize freshwater yield while controlling the supersaturation process to produce high-quality salt crystals, an approach that bridges the existing gap between desalination and salt production. This background provides the Indonesian team with the requisite expertise to spearhead the adaptation and validation of the proposed technology.

The specific challenges to be addressed by the Indonesian team include:

- **Directional Brine Management:** Developing a new design to continuously charge seawater and discharge concentrated brine for subsequent crystallization.
- **High Ambient Humidity:** Investigating the impact of reduced vapor pressure gradients on evaporation rates and overall system efficiency.
- **Fluctuating Solar Insolation:** Assessing system performance and developing an adaptive design for intermittent cloud cover, which is characteristic of the tropics.
- **Long-Term Reliability:** Evaluating the system's resilience to salt creeping, efflorescence, and biofouling in a coupled desalination-crystallization setup.

6. OBJECTIVES

The primary objective of this research is to design, fabricate, and validate a robust, high-efficiency solar-powered system for the coproduction of freshwater and salt, specifically optimized for operation under tropical climate conditions.

The specific objectives are:

1. To develop and validate a **double-cell solar desalination unit** capable of continuous, directional discharge of concentrated brine.
2. To design and optimize a **multi-stage solar desalination system** with serial connections for efficient freshwater generation and hypersaline brine production.
3. To integrate and test a **hypersaline crystallizer** driven by waste condensation heat from the desalination process to achieve efficient salt production.
4. To develop and evaluate an **adaptive design** of the integrated system for robust and predictable performance under fluctuating, real-world tropical climate conditions, culminating in a long-term techno-economic analysis.

7. METHODOLOGY AND WORK PLAN

The research will be executed over 24 months through four interconnected Work Packages (WPs), which directly correspond to the project objectives and the tasks outlined in the work statement.

This partnership leverages the complementary expertise of the Indonesian and Chinese

teams to create a powerful synergy. The division of labor is clearly defined: the Chinese team from SJTU, as the inventors of the core technology, will lead the conceptual design and initial simulations (covered in activities WP1.1, WP2.1, WP3.1, WP4.1). The Indonesian team will be responsible for the manufacturing of the lab-scale prototype, design adaptation for local conditions, and conducting comprehensive performance testing under the tropical climate (covered in activities WP1.2, WP1.3, WP2.2, WP3.2, WP4.2). In the final stage, the design tested and validated by the Indonesian team will serve as the basis for large-scale fabrication in China.

Coordination will be managed through bi-weekly virtual meetings to ensure seamless communication, monitor progress, and jointly solve technical challenges. Tasks will be jointly managed using a shared project timeline and deliverable tracker. The collaboration will be sustained beyond the project through joint applications for future funding and continued co-supervision of graduate students.

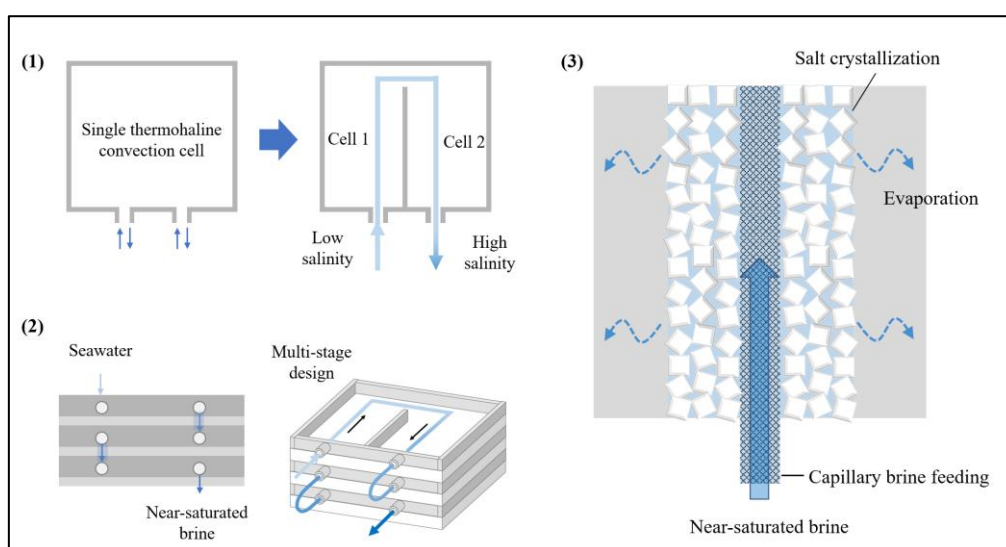


Figure 1: Conceptual schematic of the TROPIC-SALT system. The proposed research methodology involves (1) the development of a double-cell design for directional saline discharge, (2) the integration into a multi-stage architecture for hypersaline brine production, and (3) the coupling with a crystallizer for salt recovery from the near-saturated brine.

Work Package 1: Double-Cell Solar Desalination for Directional Saline Discharge This WP focuses on overcoming the limitation of previous designs by developing a system for continuous, directional flow.

- **Activity 1.1: Simulation and Design.** We will conduct numerical simulations (CFD) to model the flow pattern, temperature distribution, and salinity distribution in a novel double-cell configuration. Various connection designs between the charging and discharging cells will be evaluated to optimize directional brine flow.
- **Activity 1.2: Prototype Fabrication and Validation.** Based on simulation results, an optimized double-cell prototype will be fabricated. Experiments will be conducted to validate directional saline discharge, measuring and optimizing the salinity difference between the

inlet and outlet.

- **Activity 1.3: Performance Testing.** The prototype will be tested with varying feed salinities (from 3.5 wt% to >15 wt%) to characterize its performance under conditions representative of different stages in a multi-stage system.

Work Package 2: Multi-Stage System Design and Hypersaline Production This WP aims to scale the single-unit design into a highly efficient multi-stage system.

- **Activity 2.1: System Modeling and Optimization.** A modeling framework will be developed to analyze the multi-stage design. We will investigate the trade-offs between the number of stages, the resulting salinity gradient, and the overall thermal efficiency. The model will be used to optimize the system for maximum freshwater and hypersaline brine output.
- **Activity 2.2: Multi-Stage Prototype Fabrication and Testing.** Based on the optimization analysis, a multi-stage prototype featuring serial saline charging-discharging will be fabricated and tested in a controlled environment to validate the model predictions.

Work Package 3: Integrated Crystallizer for Salt Coproduction This WP focuses on closing the loop by using the hypersaline brine and waste heat to produce salt.

- **Activity 3.1: Crystallization Dynamics Study.** We will investigate salt creeping and efflorescence behavior on a plate heated by low-grade condensation heat from the desalination unit. This will be studied via simulation and experiments under varying salinity and heat flux conditions.
- **Activity 3.2: Fabrication and Testing of the Integrated System.** The optimized multi-stage desalination unit (from WP2) will be coupled with the crystallizer. The fully integrated system will be tested to demonstrate and quantify the simultaneous coproduction of freshwater and solid salt.

Work Package 4: Adaptive Design and Long-Term Performance Analysis This WP addresses the challenge of real-world operation in the tropics.

- **Activity 4.1: Dynamic Modeling and Adaptive Design.** A dynamic model of the integrated system will be developed to simulate its performance under fluctuating solar irradiation. This model will inform an adaptive design optimized for real tropical weather conditions.
- **Activity 4.2: Outdoor Experimentation and Performance Prediction.** The adaptive prototype will be tested outdoors to gather performance data under real-world conditions. A performance prediction model will be developed based on the experimental results.
- **Activity 4.3: Long-Term Performance and Techno-Economic Analysis.** Using the prediction model and historical tropical climate data, a long-term (annual) analysis of water/salt production and operational cost will be conducted to assess the technology's viability.

Timeline and Milestones

Activity	Y1 Q1	Y1 Q2	Y1 Q3	Y1 Q4	Y2 Q1	Y2 Q2	Y2 Q3	Y2 Q4	Remarks / Milestones
Kick-off Meeting	X								Team alignment, final design review
WP 1: Double-Cell Unit									
1.1 Simulation & Design	X	X							Optimized double-cell design completed
1.2-1.3 Fabrication & Test			X	X					WP1 prototype validated
WP 2: Multi-Stage System									
2.1 Modeling & Optimization				X	X				Multi-stage design optimized
2.2 Fabrication & Test						X	X		WP2 prototype validated
WP 3: Integrated Crystallizer									
3.1 Crystallization Study					X	X			Crystallization dynamics understood
3.2 Integrated System Test							X	X	WP3 integrated system validated
WP 4: Adaptive Design									
4.1 Dynamic Modeling						X	X		Adaptive design completed
4.2 Outdoor Testing								X	Outdoor performance data gathered
4.3 Final Analysis & Report				X				X	Final report, publications submitted
Researcher Exchange			X			X			2 researchers per side exchanged

8. EXPECTED OUTPUTS AND OUTCOMES

Outputs:

1. A fully functional, laboratory-validated prototype of the integrated TROPIC-SALT system.
2. A minimum of two co-authored publications in reputable, peer-reviewed international journals, fulfilling a mandatory requirement of the call.
3. A comprehensive dataset on the performance of advanced solar desalination technology under tropical conditions.
4. A potential joint patent application for the adaptation of the technology to tropical environments.

Outcomes:

1. Enhanced research and innovation capacity in Indonesia within the strategic field of solar energy and desalination technology.
2. Establishment of a sustainable, long-term research collaboration between the Indonesian PI's institution and Shanghai Jiao Tong University.
3. Provision of a scientific and technological foundation for developing low-cost, decentralized, and environmentally sustainable freshwater and salt production systems for Indonesian coastal communities.
4. Contribution to the national goals of energy transition and resource security.

9. COLLABORATION MECHANISM AND PARTNERSHIP OVERVIEW

This partnership leverages the complementary expertise of the Indonesian and Chinese teams to create a powerful synergy. The SJTU team, as the inventors of the core TSMD and HPSE technologies, brings unparalleled knowledge of the fundamental principles, theoretical modeling, and optimal design strategies. This is a new partnership, established through mutual scientific interest in advancing solar desalination technology for real-world applications. The Indonesian team brings essential expertise in tropical environmental conditions, local material sourcing, and practical implementation challenges, including biofouling.

Coordination will be managed through bi-weekly virtual meetings via shared platforms to ensure seamless communication, monitor progress, and jointly solve technical challenges. Tasks will be jointly managed using a shared project timeline and deliverable tracker. The collaboration will be sustained beyond the project through joint applications for future funding and continued co-supervision of graduate students.

10. BUDGET SUMMARY (*Refer To BRIN Templates Separately*)

The total proposed budget for this two-year research project is IDR 1,988,900,000. This funding is allocated as IDR 997,850,000 for the first year and IDR 991,050,000 for the second year. The budget is structured to comprehensively support all proposed research activities, with primary cost components including: prototype fabrication and component procurement, which covers raw materials like aluminum sheets and PTFE membranes as well as essential equipment such as a CDAQ data acquisition system and various sensors ; extensive testing and analysis, which includes annual licensing for CFD software and fees for specialized characterization services (SEM, ICP-MS, XRD) through ELSA BRIN ; and long-term field activities, which allocates funds for transportation, accommodation, and honoraria for field assistants. Additionally, a separate funding request for the Researcher Mobility program, managed by the BRIN Directorate of Talent Management, has been proposed at IDR 192,800,000 per year to facilitate research visits to Shanghai Jiao Tong University.

11. ETHICS, PERMITS, AND ADMINISTRATIVE CONSIDERATIONS

The research teams at Universitas Indonesia (UI) and Shanghai Jiao Tong University (SJTU) will adhere to all institutional and national regulations governing research ethics, safety, and

administration. As this project involves no human or animal subjects, specific ethical clearance is not anticipated. The Principal Investigators will be responsible for securing all necessary administrative approvals, including any required foreign researcher permits, and will establish a formal agreement on data management and intellectual property rights at the project's inception. All activities and reporting will comply with the standards set forth by BRIN and MOST.

12.RISK MANAGEMENT AND SUSTAINABILITY PLAN

Potential Risk	Likelihood	Impact	Mitigation Strategy
Technical: Lower-than-expected performance due to high humidity.	Medium	High	The system design will be iteratively optimized. The multi-stage TSMD is inherently robust, and the HPSE's high evaporation rate is expected to counteract humidity effects.
Operational: Severe and rapid biofouling clogs the system.	Medium	High	The project includes a dedicated work package to study and mitigate biofouling. The self-cleaning nature of the HPSE is a primary mitigation factor. Anti-fouling coatings will be explored as a secondary measure.
Logistical: Delays in sourcing materials for fabrication.	Low	Medium	Prioritize locally available, low-cost materials in the design phase. Establish contact with multiple suppliers early in the project.

13.REFERENCES

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[2] Z. Xu *et al.*, "Ultrahigh-efficiency desalination via a thermally-localized multistage solar still," *Energy Environ. Sci.*, vol. 13, no. 3, pp. 830–839, 2020.

[3] World Bank, *Indonesia: Climate Risk Profile*, 2021 and Bappenas, *Status Lingkungan Hidup Indonesia 2022*, 2022.

[4] Pusat Penelitian, Badan Keahlian, Sekretariat Jenderal DPR RI, *Transforming Indonesia's Salt Industry: From Import Dependence to Self-Sufficiency*, 2024.

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[6] J. Gao *et al.*, "Extreme salt-resisting multistage solar distillation with thermohaline convection," *Joule*, vol. 7, no. 10, pp. 2274–2290, Oct. 2023.

- [7] J. Yu *et al.*, "Self-assembled porous salt crystals for solar-powered crystallization," *Energy Environ. Sci.*, vol. 18, pp. 454-467, 2025.
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- [9] L. Putriyana, M. Nuriyadi, E. Djubaedah, Y. Gunawan, and N. Nasruddin, "Investigating Techno-Economic Feasibility of Geothermal Polygeneration in Nusalaut Island, Central Maluku District, Maluku Province," *EVERGREEN Joint Journal of Novel Carbon Resource Sciences & Green Asia Strategy*, vol. 10, no. 4, pp. 2464-2475, 2023.
- [10] A. S. Ekariansyah, M. Muwonge, M. R. Saefuttamam, Y. Dikaimana, and Nasruddin, "Energy, exergy, and exergy-economic optimization of a multigeneration system driven by geothermal primary heat source using multi-objective genetic algorithm (MOGA)," *Energy*, vol. 330, p. 136653, 2025.

14. ANNEXES

Annex 1: Budget Table (RAB)

Annex 2: Curriculum Vitae (CV)

Annex 3: Agreement on Joint Research